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RESEARCH ARTICLE

Student Performance Prediction Approach Based on Educational Data Mining

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ABSTRACT Predicting student performance is crucial for improving students' future academic achievements. Within student groups, common characteristics can reveal trends in overall student learning. Most studies tend to focus on the common characteristics of students but ignore their individual characteristics. However, individual characteristics are important in promoting student academic performance because they allow us to understand the unique learning performances of each student. To address this issue, this paper proposes a student performance prediction approach. First, addressing the problem of difficulty in effectively dividing the student samples under multi-dimensional discrete data, we propose a method that combines the relationship matrix-based bipartite network approach (RMBN) with Louvain clustering. Second, the hybrid neural network model based on a relationship matrix (RMHNN) is proposed to address the problem that discrete types of features are difficult to fit by algorithms. The results show that the implementation of the model on real student data can effectively predict student performance with an accuracy of 93.1% and an F1-score of 90.45%. With the model's predicted student performance, educators can provide individualized support and assistance to each student.

INDEX TERMS Performance prediction, data mining, common characteristics, individual characteristics, relation network.

I. INTRODUCTION

Academic performance is the most important measure of the quality of college students, serving as the fundamental criterion for universities to monitor teaching quality and evaluate students' academic abilities [1]. In school period, student's performance is reflected in both academic and practical performance. Academic performance refers to students' performance in various courses, while practical performance includes students' performance in research competitions and on-campus scholarships, among other aspects. Students in the same college or university have similar test scores at the time of enrollment. However, a variety of factors can lead to a gradual widening of performance between students during their studies. Numerous researchers have conducted extensive studies to address the issues of incomplete assessment of student performance and the lack of tools for assessing students' performance in most universities.

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Numerous studies related to student development have been published within the field of education. Numerous studies cover various aspects, including student dropout rates [2], student profiling [3], [4], [5], monitoring student academic performance [6], [7], [8], [9], and predicting at-risk students [10]. In research concerning student dropout rates, students are typically classified as either "good" or "poor" [11], [12], [13], with targeted interventions aimed at high-risk student groups. Student profiling studies [14] involve the creation of student profiles by collecting behavioral and academic data to enable early detection of students exhibiting abnormal behavior. Additionally, there have been endeavors to predict groups of students facing learning difficulties based on their assignment submissions [15], as well as efforts to identify students' learning styles through the analysis of their academic data characteristics [16].

Most of the research on predicting student performance focuses on extracting common characteristics among student groups while neglecting the consideration of individual characteristics of students. There are three problems with

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these studies. Firstly, in terms of the choice of data objectives [17], [18], most studies choose to use students' overall grades to assess students' academic performance. However, this approach ignores the performance of each student in different subjects, making it difficult to extract individual characteristics. Secondly, in terms of the choice of research method, most of the studies employ a supervised learning approach [19]. Since assessing student learning is a complex concept, it is difficult for us to accurately measure the problem with a single target variable. Using an unsupervised learning approach can more fully utilize all the data features to mine the individual differences between students. Thirdly, in terms of student group division [11], [15], [20], most studies tend to categorize students as ''good'' or ''poor''. While this categorization helps provide early warnings for students classified as ''poor'', it is overly broad for students classified as ''good'', potentially overlooking individual differences within this category.

This paper focuses on both individual and common characteristics of students, aiming to predict students' academic performance by exploring these characteristics. Firstly, to address the problem of k-means struggling to effectively cluster student samples in multi-dimensional discrete data, we propose a method that combines the relationship matrix-based bipartite network approach (RMBN) with Louvain. Secondly, to tackle the issue that existing studies using a single objective variable may not accurately assess student performance, we use a multi-objective assessment method. We propose RMBN approach. Thirdly, to address the problem that discrete features are unsuitable for learning within a feature space based on Euclidean distance, we propose the hybrid neural network model based on a relationship matrix (RMHNN). Fourthly, to resolve the concern that existing research tends to focus on students' common characteristics but neglects their individual characteristics, we propose the student performance prediction approach.

The main contributions of the paper are summarized as follows:

- We address the difficulty of k-means in effectively classifying student samples under multi-dimensional discrete data by combining the proposed RMBN approach with Louvain.
- We use a multi-objective assessment method and propose RMBN. By using the RMBN data processing approach, we can comprehensively consider multiple performance indicators, addressing the difficulty of capturing associations between sample features when analyzing multidimensional data, which helps us achieve the goal of accurately assessing student performance.
- We propose the RMHNN model to solve the problem of discrete types of features that are difficult to fit. Experimental results show that the model can accurately predict student performance.
- We apply the proposed a student performance prediction approach to real student data and compare it with traditional machine learning algorithms and neural networks. The experimental results show that our model is valid.

The remaining sections of this paper are organized as follows. Section II presents the related work. Section III provides an overview of data collection, data integration, and data cleaning processes for the dataset. Section IV offers a detailed explanation of the solution design for predicting students' future academic performance. Section V evaluates the performance of the adopted methods and presents the results analysis. Finally, section VI concludes the paper.

II. RELATED WORK

With the rapid growth of student data and the continuous development of algorithm models, we are now able to construct more effective student prediction models. One common approach is to use a grading mechanism for categorization. Nghe et al. [17] divided students into categories based on grade point average (GPA) and used students' multidimensional data features and yearly academic GPA to predict their GPA category for the next academic year. Additionally, depending on the research needs and student characteristics, different methods can be used for student categorization and prediction. Akram et al. [15] aimed to predict students with learning difficulties based on their assignment submission behavior. They used the k-means clustering algorithm to label students as procrastinators and non-procrastinators, followed by applying different classification methods to categorize the students. By integrating behavioral categorization models with behavioral data, they obtained category-specific feature values for each behavior, enabling the construction of machine learning-based performance prediction. Asif et al. [20] conducted research based on college students' performance and divided the student group into low-achieving and high-achieving for nurturing and intervention purposes.

In the application of student performance prediction, most studies have employed machine learning algorithms. Yağcı [21] proposed a new model based on machine learning algorithms to predict undergraduates' final exam scores using midterm exam scores as the source data. The performance of the proposed model was compared with other machine learning algorithms, and the results showed a classification accuracy of 70%-75%. In terms of student dropout rates, Niyogisubizo et al. [22] used an ensemble model to predict students' dropout status in university courses. When it comes to students' choices of elective courses, Huang et al. [23] developed a score-based elective course recommendation system that accurately predicted the course scores for each student by leveraging the course grade distributions of similar senior students. Tomasevic et al. [11], through increasing the available data sources gradually and employing state-of-the-art supervised machine learning techniques to perform a comprehensive analysis and comparison. They discovered that increasing the available data improves the detection performance of educational data mining. To identify the

importance of factors influencing student performance, Alshanqiti and Namoun [24] proposed a hybrid regression model and multi-label classifier to explore the various factors influencing student grades. In addition to machine learning, some studies have also utilized deep learning methods. In contrast to the majority of existing research that utilized machine learning to predict student performance, Akour et al. [25] addressed and validated the prediction of student grades using a convolutional neural network-based deep learning algorithm.

Although there have been many studies using data mining techniques to predict student performance, most of them have focused on mining the common characteristics of the student groups and lacked a deeper dive into the individual characteristics of each student. In the student performance prediction task, this paper considers both the common characteristics of the student body as a whole as well as the individual characteristics of students in different subgroups.

III. DATA

A. STUDENT DATASET

The availability of data and the feasibility of the research are fundamental aspects of conducting scientific studies. As shown in Table 1, we utilizes student data from all majors within the information school at a university for 2017 and 2018.

B. DATA PREPROCESSING

We extract the students' names and course GPAs from five data tables, integrated them using students' names as the key, and remove duplicate data. Additionally, we standardize synonymous attribute fields in the dataset to a uniform format and retain only one attribute feature. Sensitive information in the data tables is anonymized.

Since the cleaned dataset contains both discrete and continuous features, we apply a binning strategy to the continuous course grade data [26], [27]. This process transforms continuous data into discrete categories. Figure 1 illustrates the mapping relationship for discretizing course grades, where continuous GPAs are segmented into five distinct categories. Moreover, we use one-hot encoding for categorical data types, representing each category with a binary vector. Additionally, character data is converted into integer data types.

Finally, the grade matrix S is generated, which comprises 20 feature attributes, including grade level (Grade), major (Major), ideological and political education (Pol.), foreign language (Fore.), physical education (Phys.), basic sciences (Math.), KAB entrepreneurship foundation (KAB), college chinese (Chi.), health guidance (Health), disciplinary majors (Core), core majors (Prof.), extended majors (Exp.), composite majors (Com.), special design (Spec.), graduation design (Grad. Proj), gender, post-graduation plans, political affiliation, school awards (No-JSA, With-JSA), and research awards (No research competition, With research competition).

Feature element ID	3-15	1-2、16-18	19-20
Feature type	Course score	Personal attribute	Award status
Feature value range	[1,5]	[0-N]	[0,N]
Feature value description	1: Fail 2: Pass 3: Average 4: Good 5: Excellent	0-N scaled to 0-N	0-N scaled to 0-N

FIGURE 1. Structure of the analyzed data (student-related features).

IV. METHODOLOGY

A. APPROACHES TO PREDICTING STUDENT PERFORMANCE

We propose the student performance prediction approach. As shown in Figure 2. To comprehensively assess student performance, we collect student performance characteristics at both the academic and practical performance levels to enable the construction of a dataset that can accurately reflect student performance. The integrated and cleaned data includes both discrete-type features and continuous-type features. In the case of discrete-type features, the distribution of eigenvalues is too sparse. During data preprocessing, we transform continuous features into discrete ones using data binning and convert character-type features into discrete features through one-hot coding. After the data preprocessing is complete, we obtain a grade matrix S with too large a feature space and too sparse a feature representation vector. Because the eigenvalues of the distribution in the grade matrix S exhibit high dimensionality and discrete properties, we have difficulty capturing the associations between students from this data. First, we propose the RMBN data processing approach to address the problem of weak correlation between students in a high-dimensional and discrete grade matrix S. The RMBN data processing approach projects the data into a bipartite network G after encoding the entire multidimensional data in one-hot. With the data structure of this graph network, the associations between students and variables can be made explicit, which helps subsequent algorithms capture similarities and differences between features. Second, the Louvain clustering algorithm can fully take into account the associations between students in complex graph networks. We use Louvain clustering as a graph-based feature extraction method for bipartite network G to form grouping results with strong similarities between students within groups and strong differences between students outside groups. Third, after obtaining metrics to accurately assess student performance, we research the prediction of student performance. In response to the fact that the feature values in the grade matrix S are in the form of high-dimensional and discrete data, and the discrete features are not suitable for learning within a feature space based on Euclidean distance, we propose the RMHNN model to predict students' academic performance.

TABLE 1. Data sources and dataset variables.

Table Name	Variables
academic performance dataset	student course name, student course GPA, student course credits, course retake, course regular assessment grade, course final exam grade, course makeup exam grade.
basic student information dataset	student age, student gender, student political affiliation, student contact number.
scholarship during enrollment dataset	includes the name of the scholarship received by the student on campus and the time when the student received the on-campus scholarship.
research competition results dataset	includes the name of the research competition, the time of the award, and the names of the student team members.
post-graduation destinations dataset	contains information about students' post-graduation destinations and their employment nature.

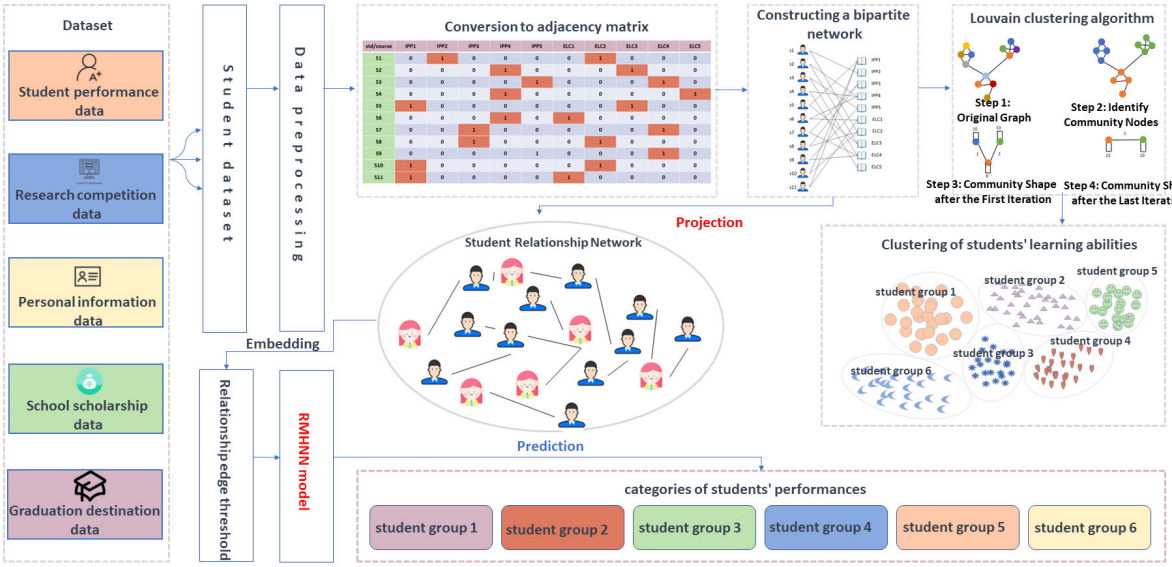


FIGURE 2. Student performance prediction approach based on educational data mining.

B. THE RMBN DATA PROCESSING APPROACH

The feature space of the grade matrix S obtained after data preprocessing is too large and the feature representation vector is too sparse. When processing this data using traditional machine learning algorithms and neural networks, it is difficult to capture the association of eigenvalues between samples. We propose the RMBN data processing approach. The RMBN data processing approach undergoes two processing steps. First, we encode the entire preprocessed dataset with one-hot coding to generate a relation matrix with only 0 and 1 eigenvalues. Doing so helps to highlight patterns and structures in the data and clarify the links between samples and variables in the grade matrix S. In addition, numerical differences between categories in the grade matrix S can be eliminated by this treatment, so that subsequent algorithms do not incorrectly assign greater weights to certain categories because of the magnitude of the values, realizing the goal of coding each category with the same weight. Second, to solve the problem that the data feature space becomes too large and the feature representation vector becomes too sparse after one-hot coding, we project the relation matrix as a bipartite network [28], [29], [30], [31], and this treatment helps to clarify the structure of relations in the data.

C. THE LOUVAIN CLUSTERING ALGORITHM

In complex graph networks, the Louvain clustering algorithm can comprehensively take into account the associations between students [32], [33], [34], which in turn results in groupings of high similarity between students within groups and significant differences between students between groups. Therefore, this paper adopts Louvain clustering as a graph-based feature extraction method to capture both common characteristics of student groups and individual characteristics of each student within high-dimensional and discrete data.

D. THE RMHNN MODEL

As shown in Figure 3. The RMHNN model includes an input layer, a graph convolution layer, and a cross-attention layer. First, the grade matrix S and the student relation network A are input to the input layer. Second, the graph convolutional

network layer embeds the relation network A into the grade matrix S. By setting the threshold range of relation edges and other methods, it helps the model maximize the learning of the hidden feature association information in the graph network, thus forming a new matrix representation. Third, the cross-attention layer helps to assign weights to factors that affect student performance. Finally, the linear layer is utilized to fuse the newly formed data features to complete the prediction of student performance.

To capture correlations in student academic performance, we design a method based on the concept of social network and according to our defined rule that "if students have similar course grades under the same course, there is a dependency relationship between these two students, and the weight of the undirected edges connecting these two students' nodes is 1". Under the rule, we design the single graph extraction algorithm based on node connectivity relationships (BNSN) that can extract the student nodes and the undirected edges between them from the bipartite network G generated based on the RMBN data processing approach, and calculate the weights of these undirected edges. The pseudocode for the BNSN(G) algorithm1 is as follows:

Algorithm 1 BNSN(G) Algorithm

input : Bipartite network G
output: Relation network A

Convert the bipartite network G into a triplet
$T = (U, V, E)$;
for $i = 0$ **to** $T.size[0]$ **do**
  Grouping of students according to their node number;
end
C=[];
for $i = 0$ **to** $T.size[0]$-1 **do**
  Q=[];
  initialization $count$;
  Extract all variables v owned by the student with node i ;
  $V_i = \{ v_{i1}, v_{i2}, v_{i3}, \dots \}$;
  for $j = i+1$ **to** $T.size[0]$ **do**
    $count$=0;
    Extract all variables v owned by the student with node j ;
    $V_j = \{ v_{j1}, v_{j2}, v_{j3}, \dots \}$;
    if $X \in V_i \cap V_j$ **then** $count$++;
    else $count$=0;
    Q.append(i,j,$count$);
  end
  C.append(Q);
end

In the input layer of the RMHNN model, we use the student relation network A with the grade matrix S as the student characteristics for the student performance prediction task. Based on the Louvain grouping scheme, we categorize each student's academic performance into n categories, student group 1, student group 2, etc., which are denoted as $target = \{1, \dots, n\}$.

In the RMHNN model, the graph convolution layer utilizes graph convolutional network (GCN) [35], [36] to embed the student relation network into the student data, enriching the representation information of the student data and better capturing the mutual influence and network social structure among students. In the graph convolutional layer, we optimize the adjacency matrix in the GCN by changing the threshold of relationship edges, using a linear scaling function to resize the adjacency matrix, and other methods to improve model performance. Furthermore, we also conduct normalization and adjust the feature dimensions.

In the RMHNN model, the cross-attention layer utilizes a cross-attention network [37], [38] to handle the correlation between different features, better understanding the importance and influence among different features. In the cross-attention layer, we enhance the model's predictive performance by integrating various data sources to explore the best dataset combinations.

E. EVALUATION METRICS

Modularity(Q) is an important metric used to measure the density of community structure distribution [39]. The formula for modularity(Q) [40] is as follows:

$$Q = \frac{1}{2m}\sum_{i=1}^{N}\sum_{j=1}^{N}(A_{i,j} - \frac{d_i d_j}{2m})\delta(c_i, c_j) \tag{1}$$

In equation (1), the undirected graph formula, N is the network size, $A_{i,j}$ represents the adjacency matrix between node i and node j. The variable $2m$ represents the sum of all edges in the adjacency matrix A. d_i represents the degree of node i, d_j represents the degree of node j. c_i represents the community cluster to which node i belongs, and c_j represents the community cluster to which node j belongs, and the δ function takes if node i and node j are in the same community, $\delta(c_i, c_j)$ is equal to 1; otherwise, it is equal to 0.

Accuracy ($y_{accuracy}$) is one of the commonly used evaluation metrics in classification models. Precision ($y_{precision}$) is one of the evaluation metrics for classification models. Recall (y_{recall}) is one of the evaluation metrics for classification models. F1-score (y_{F1}) is a comprehensive evaluation metric. Their formulas are as follows:

$$y_{accuracy} = \frac{y_{TP} + y_{TN}}{(y_{TP} + y_{FP} + y_{TN} + y_{FN})} \tag{2}$$

$$y_{precision} = \frac{y_{TP}}{(y_{TP} + y_{FP})} \tag{3}$$

$$y_{recall} = \frac{y_{TP}}{(y_{TP} + y_{FN})} \tag{4}$$

$$y_{F1} = 2 \times (\frac{y_{precision} \times y_{recall}}{(y_{precision} + y_{recall})}) \tag{5}$$

In equation (2-5), y_{TP} represents the number of samples where both the actual and predicted labels are positive,

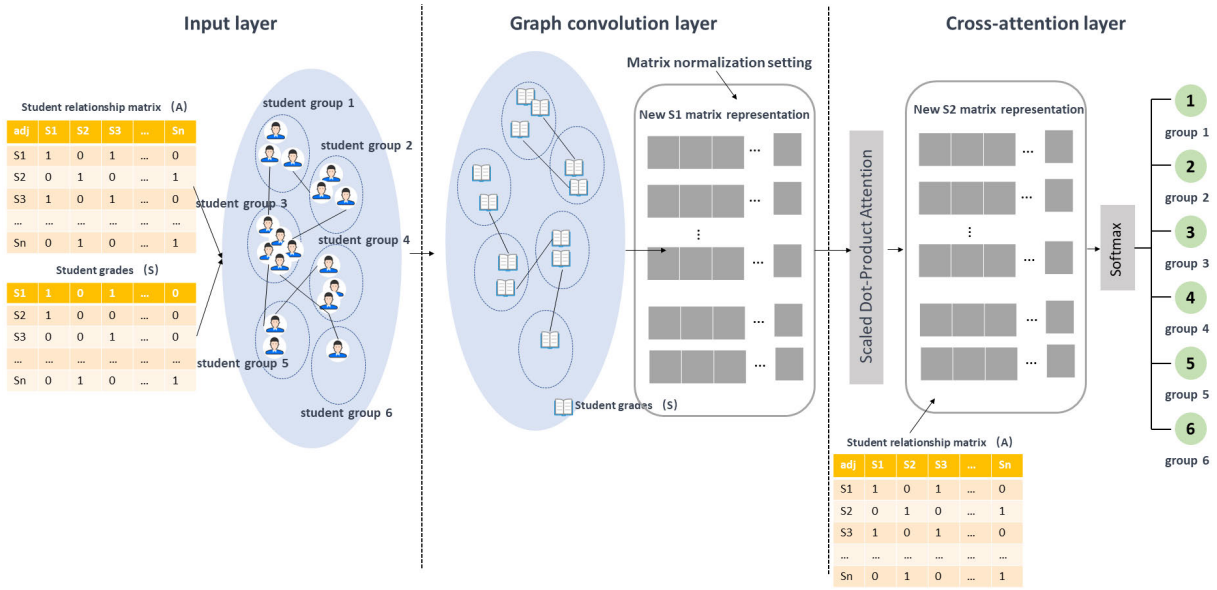


FIGURE 3. The hybrid neural network model based on a relationship matrix (RMHNN).

indicating consistent predictions with the actual labels. y_{TN} represents the number of samples where both the actual and predicted labels are negative, indicating consistent predictions with the actual labels. y_{FP} represents the number of samples where the actual label is negative and the predicted label is positive, indicating inconsistent predictions. y_{FN} represents the number of samples where the actual label is positive but the classifier model predicts it as negative.

V. RESULTS AND DISCUSSION

A. ANALYSIS OF THE STUDENT PERFORMANCE PREDICTION MODEL

First, in the task of evaluating student performance, to assess student performance, literature [5], [8], [16], [20] uses clustering algorithms to group similar or related objects together. The k-means clustering algorithm is often used to deal with continuous data The k-means clustering algorithm is often used for continuous data because it clusters based on the distances between samples, and distance measures are typically meaningful for continuous data [41]. The dataset used in this paper contains both discrete-type features and continuous-type features, and after data preprocessing, the resulting grade matrix S is all discrete-type features. Therefore, the k-means algorithm may be less suitable for the dataset used in this paper. To address the challenge of clustering multiple discrete features accurately, we combine the proposed RMBN data processing approach with Louvain clustering. The experimental results demonstrate that this combination reduces the dimensionality of the encoded data, clarifies the connections between students and variables, and achieves highly modular grouping results.

Second, in the task of predicting student performance, literature [16] uses decision trees (DT) as classifiers. Literature [42] uses random forest, support vector machines (SVM), logistic regression, plain bayes, and k-nearest neighbor (KNN) from machine learning algorithms as classifiers. Literature [21] uses a convolutional neural network (CNN) as a classifier. Literature [43] uses an attention-based graph convolutional network as a classifier. To achieve accurate student performance prediction, we introduce the RMHNN model. To validate the effectiveness of our proposed RMHNN model, we apply the classification algorithms from previous studies to our data with substitutions. The experimental results demonstrate that our proposed model surpasses other classification algorithms in accuracy and can effectively predict student performance.

Finally, for student performance assessment and student performance prediction, we propose a combination of the RMBN data processing approach and Louvain clustering for assessed student performance, and the RMHNN model for predicted student performance, respectively. The experimental results show that our proposed a student performance prediction approach is effective.

B. LOUVAIN ALGORITHM CLUSTERING RESULTS

In the Louvain clustering algorithm, resolution is a parameter used to adjust the quantity of community nodes. In Table 2, when the resolution is set to 1, Newman*, Dugue*, and Potts* use data processed based on the RMBN data processing approach. The purpose of this approach is to convert the preprocessed student dataset into a bipartite network, where the values of the relationship edge connecting the student nodes to the variable nodes in the formed graph network differ due to the inconsistency of the eigenvalues between the different features. This may lead to subsequent algorithms confusing the correlations between students. Therefore, to balance the weights of the relationship edges in the graph network, this paper encodes the dataset once again with

one-hot coding to make the connection between the student nodes and the variable nodes in the graph network clearer. Newman, Dugue, and Potts used standardized processed data. Based on the modularity values (Q), it can be observed that the clustering results in the left three columns have higher Q, namely 73.37%, 71.37%, and 19.30%, respectively. On the other hand, the clustering results in the right three columns have lower Q, namely 4.39%, 48.99%, and 6.28%, respectively. The clustering results in the left three columns yield 6, 7, and 72 communities, while the results in the right three columns yield 10, 23, and 50 communities, respectively. According to the modularity evaluation metric, the clustering results in the left three columns exhibit higher Q, indicating a higher density of connections within communities compared to connections between communities, and thus a tighter community structure. The experimental results indicate that after the data processed by the RMBN data processing approach, the Louvain clustering algorithm can effectively identify the community structure within the network and assign student nodes to their respective communities. Based on the experimental results, it can be observed that after the data is processed by the RMBN data processing approach, and then applying the Louvain clustering method for grouping yields better experimental results compared to directly using Louvain clustering for grouping the preprocessed dataset after standardization. This is due to the fact that one-hot coding is taken to discretize the features throughout the RMBN data processing approach, where each feature is transformed into a binary feature. Because each feature is turned into an explicit node, this explicit feature representation helps Louvain clustering to capture similarities and differences between features more easily.

In this paper, we use the Louvain (Newman*) clustering algorithm to group the data processed through the RMBN data processing approach and then conduct a visual analysis of the generated six student groups. In the visual results, these six student groups exhibit distinct trends in their academic performance. As shown in Figure 4 and Figure 5. Students in student group 2 excel academically and in practice. The student in student group 5 needs to be strengthened in terms of academics and in terms of practice. Students in student group 5 need to enhance their theoretical knowledge in academics and participate in research competitions to improve their practical skills. Students in student group 1, student group 3, student group 4, and student group 5 need to strengthen their academic performance in courses related to mathematical fundamentals and subject-specific foundational courses. By examining the academic and practical performance of students in different programs, we can see that there is room for improvement for all student groups.

C. RESULTS OF THE RMHNN MODEL

In the experiment, we adjust the relationship edge threshold range of the adjacency matrix to observe its impact on the model's predictions. As shown in Figure 6, when no threshold

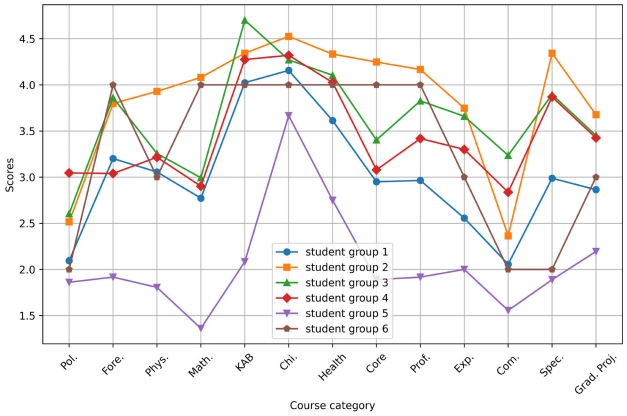


FIGURE 4. Visualization of student academic performance in different groups (student academic performance and student practical performance).



FIGURE 5. Heatmaps of different student groups' performance in extracurricular practices.

is set, the GCN achieves remarkably low accuracy and F1-score, with values of only 7.07% and 2.64%, respectively, while the graph attention network (GAT) performes slightly better with an accuracy of 34.48% and an F1-score of 17.68%. This indicates that both models struggle to classify the student data without a threshold. As the threshold for the connection strength increases, the accuracy and F1-score of both the GCN and GAT gradually improve. When the threshold value of the relationship edge is set to be greater than 8, the accuracy and F1-score of the GCN significantly improve to 63.64% and 48.65%, respectively. Similarly, the accuracy and F1-score of the GAT also see significant improvements, reaching 77.01% and 77.51%, respectively. As the relational edge thresholds are adjusted, the prediction accuracies and F1-score of both the GCN and GAT stabilize. Specifically, when the threshold value of the relationship edge is set to be greater than 14, the GCN achieves its highest prediction accuracy and F1-score at 84.85% and 63.64%, respectively. For the GAT, the peak prediction accuracy and F1-score are reached when the threshold value of the relationship edge is set to be greater than 15, at 81.61% and 81.02%, respectively. These experiments demonstrate that adjusting the adjacency matrix by setting a threshold range for relationship edges effectively enhances the model's predictive performance. When the threshold value of the relationship edge is set low, there is difficulty in filtering out weaker connections between

TABLE 2. Comparative analysis of experimental results on module degree value *Q* and the number of community partition count.

Evaluation	Newman*	Dugue*	Potts*	Newman	Dugue	Potts
modularity (Q) /%	73.37	71.37	19.30	4.39	48.99	6.28
community count	6	7	72	10	23	50

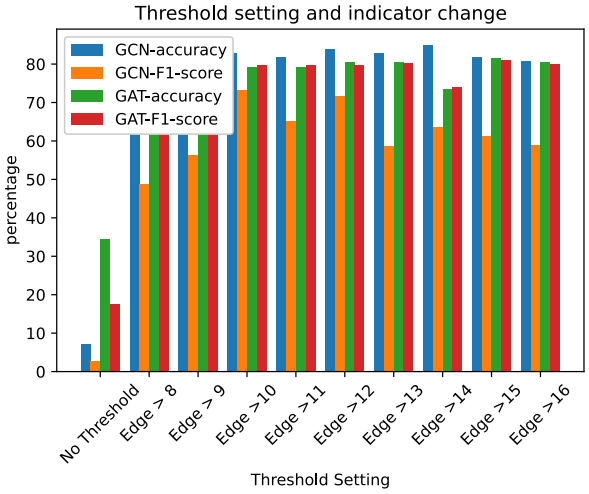


FIGURE 6. The predictive effectiveness of the model is tested by setting a threshold range to adjust the adjacency matrix(%).

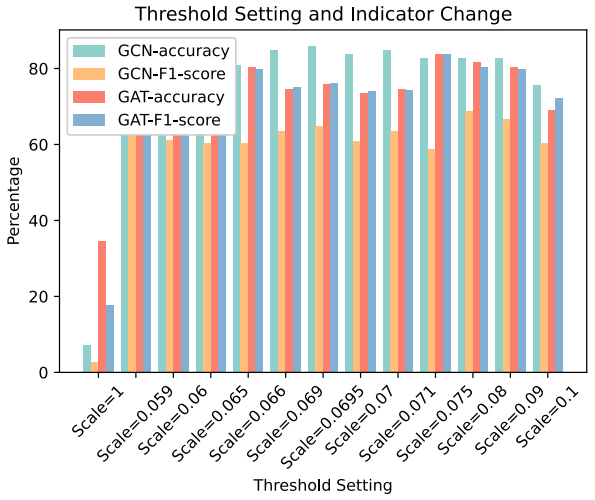


FIGURE 7. Adjusting the adjacency matrix by setting a scaling factor using a linear scaling function to evaluate the model's predictions(%).

nodes, leading to the misinterpretation of unrelated nodes as being connected. Conversely, when the threshold value is set high, it's easy to filter out stronger connections, resulting in the misinterpretation of related nodes as not being connected.

The method of adjusting the adjacency matrix by setting a threshold for the relationship edges suffers from the problem of not being able to capture subtle changes between the weights, which may lead to loss of information and affect the performance of the model. The use of a linear scaling function as the weight adjustment function can better preserve the information between the weights [44]. As shown in Figure 7, in the GCN, the accuracy and F1-score of the model peak at scale=0.0695 with values of 85.86% and 64.72%, respectively. In the GAT, the accuracy and F1-score of the model peaked at scale=0.075 with values of 83.991% and 83.79%, respectively. The experimental results show that using the linear scaling function as a weight function to scale the data matrix helps improve the performance and generalization of the model, enhancing its ability to express complex relationships.

In the GCN, the student relation network presents a certain focus, while the attention mechanism in cross-attention networks can assign weights according to the degree of importance of the features, calculate the weights of each student's performance features, and highlight the key factors affecting the student's performance, which can solve the specific weighting problem. Therefore, a cross-attention layer is introduced to balance the weights of the features affecting student performance. Figure 6 illustrates the model performance in the GCN by adjusting the relational edge thresholds to observe the model performance. Figure 8 illustrates the effect of normalized and un-normalized on the model performance with the relational edge threshold set. Under the premise of using the method of adjusting the threshold range of relationship edges, comparing the experimental results in Figure 6 with the experiments in Figure 8 where normalization is not applied, we can observe the impact of introducing the cross-attention layer on the model's prediction accuracy and F1-score. In Figure 6, the F1-score of the model reaches its highest value when the threshold of the relationship edge is set to be greater than 12, with a value of 71.62%. The highest accuracy of the model is achieved when the threshold value of the relationship edge is set to be greater than 14, with a value of 84.85%. In Figure 8, the F1-score of the model reaches its highest value with a value of 81.06% when the threshold of the relationship edge is set to be greater than 11 without normalization. The highest accuracy of the model, 80.98%, is achieved when the threshold of the relationship edge is set to greater than 12.

Figure 7 shows how the data matrix is scaled in the GCN by using a linear scaling function and observing how it affects the model performance. Figure 9 illustrates the effect of normalized and un-normalized on the model performance with the application of a linear scaling function. Comparing the experimental results in Figure 7 with those without normalization in Figure 9 under the premise of using a linear scaling function to scale the data matrix approach, we can observe the effect of introducing the cross-attention

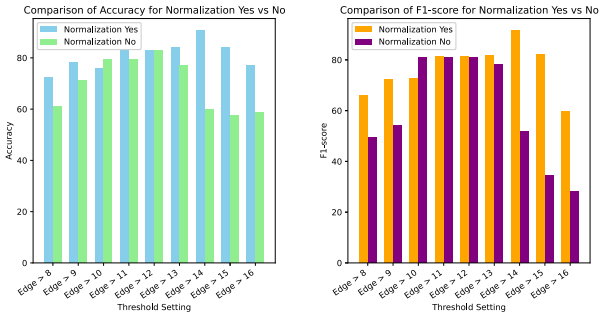


FIGURE 8. Validation of the effect of normalization on model performance in the context of the method of setting a threshold range to adjust the adjacency matrix(%).

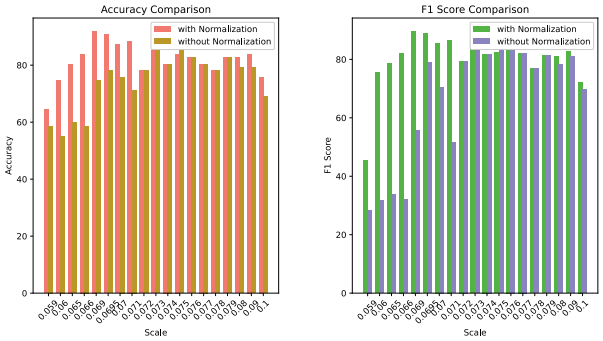


FIGURE 9. Validating the effect of normalization on model performance in a method that uses a linear scaling function to set the scaling factor to adjust the adjacency matrix(%).

layer on the model prediction accuracy and F1-score. In Figure 7, the highest accuracy of the model is achieved when scale=0.0695 and its value is 85.85%. The F1-score of the model is highest when scale=0.08 and its value is 68.84%. In Figure 9, the accuracy and F1-score of the model peaks at scale=0.075 with values of 86.20% and 87.94%, respectively, under the normalization condition. These results indicate that the introduction of the cross-attention layer can indeed improve the F1-score of the model by considering differences in how various characteristics affect different students. The cross-attention network helps address these variations in feature importance.

Normalization is a data preprocessing technique that aims to standardize the scale of various features by mapping the data to a specific range [45]. As shown in Figure 8, the impact of normalization and non-normalization on model performance is demonstrated under the condition of setting the relationship edge threshold. The accuracy and F1-score of the RMHNN model are highest when the relationship edge threshold is set to greater than 14, with values of 90.80% and 91.71%, respectively. The experimental results show that when optimizing the model by setting the relationship edge threshold range, performing data normalization can significantly improve the prediction of the model. In the student performance prediction task in this paper, performing data normalization can better improve model performance and help identify correlations between students.

As depicted in Figure 9, the influence of normalization and non-normalization on model performance is illustrated under the premise of applying a linear scaling function. Under the condition of normalization, when scale=0.069, the model achieves its highest accuracy and F1-score, which are 91.95% and 89.67%, respectively. In the absence of normalization, when scale=0.075, the model achieves its highest accuracy and F1-score, which are 86.20% and 87.94%, respectively. The experimental results show that the effectiveness of model prediction can be improved by using the normalization method.

As shown in Figure 10. In the graph convolutional layer, the performance of these methods in predicting student performance is evaluated by selecting different data dimension outputs, constructing student and course relation networks using the grade matrix S, and employing methods such as integrating the grade matrix, student adjacency matrix, and course adjacency matrix in the cross-attention layer. Under the condition "RMHNN (Edge > 14) (900)", the accuracy of the model is 90.80 and the F1-score is 91.71%. Under the condition "RMHNN (scale=0.069) (900)", the accuracy of the model is 91.95% and the F1-score is 89.67%. Under the condition "RMHNN (Edge > 14) (402)", the accuracy of the model is 93.10% and the F1-score is 90.46%. Under the condition "RMHNN (scale=0.069) (402)", the accuracy of the model is 93.10% and the F1-score is 90.45%. This shows that the choice of using different data dimensions output in the graph convolutional layer can improve the F1-score of the model. "Using the student matrix alone (Edge > 14) (900)" represents the condition that the threshold of the relation edge is set to greater than 14 in the graph convolution layer and the output dimension of the data is increased to 900, and we use the student performance matrix S alone in the cross-attention layer for our experiments. "Add Course Relationship Matrix (Edge > 14) (900)" means that in the graph convolutional layer, with the threshold value for the relationship edges set to greater than 14 and the data output dimension increased to 900, we conduct data fusion and experimental verification in the cross-attention layer using three tables: the student grade matrix S, the student adjacency matrix obtained from the student relation network, and the course adjacency matrix obtained from the course relation network. "Remove Student Relation Network (Edge > 14) (900)" means that in the graph convolutional layer, with the threshold value for the relationship edges set to greater than 14 and the data output dimension increases to 900, we conduct data fusion and experimental verification in the cross-attention layer using two tables: the student grade matrix S and the course adjacency matrix obtained from the course relation network. The experimental results show that using the student adjacency matrix obtained based on the student relation network in the cross-attention layer can improve the model prediction. In the cross-attention

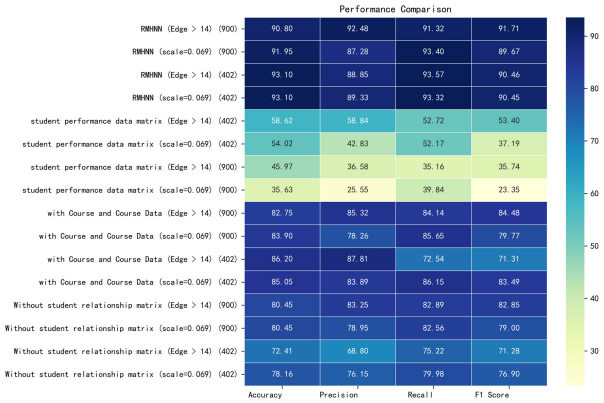


FIGURE 10. By using different data dimensions and different relational networks to assess their impact on model performance(%).

layer, fusing the new graph representation results obtained in the graph convolution layer with the student adjacency matrix, helps the model to take into account the performance characteristics of each student more accurately and improves the accuracy and F1-score of the model. Theoretically, the course relation network can improve the accuracy of the model prediction, but since the course relation network in this model is constructed based on the student performance matrix S, its use does not improve the model's accuracy and F1-score.

The results of the model comparison analysis experiments are shown in Table 3 where the RMHNN model outperforms the other models overall. The prediction accuracies of KNN, SVM, and DT were low, only 61.68%, 72%, and 66.6%. This is because traditional machine learning algorithms and neural network models are not suitable for fitting data tables where the feature space is too large and the feature representation vectors are too sparse. The SVM although predicts well in the high dimensional data space, requires more support vectors to maintain the boundaries leading to a prediction accuracy of 72% and an F1-score of only 57%. DT are able to handle mixed data types, but in high dimensional spaces, DT need to increase their depth to capture all features, but doing so raises the issue of overfitting. The prediction accuracy of the decision tree was 66.6% with an F1-score of 51%. Multilayer perceptron (MLP), backpropagation network (BPNN), and CNN are deep learning algorithms, but their prediction accuracies are only 68.97%, 67.2%, and 68.58%, due to the problem that deep neural networks are prone to overfitting in high-dimensional sparse data. The prediction results of GCN and GAT are significantly better than other models, which is because they solve the problem of weak correlation of feature values between samples under high-dimensional sparse data by means of graph network representation. They adjust the association between the samples through the embedded adjacency matrix, which allows the model to learn the dependency of the eigenvalues between the samples to improve the performance of the model prediction. The experimental

TABLE 3. Comparison of experimental results(%).

Models	Accuracy	Precision	Recall	F1-score
KNN	61.68	73.44	44	47
SVM	72	78	55	57
DT	66.6	63	51	51
MLP	68.97	80	45.40	45.54
BPNN	67.20	70.50	81.89	46.45
CNN	68.58	80.29	45.12	45.42
GCN(edge>14)	84.85	67.66	85.97	63.64
GCN(scale = 0.0695)	85.86	68.49	86.26	64.72
GAT(edge>13)	80.46	82.06	80.46	80.23
GAT(scale=0.075)	83.91	85.28	80.46	83.79
RMHNN(edge>14)(900)	90.80	92.48	91.32	91.71
RMHNN(scale=0.069)(900)	91.95	87.28	93.40	89.67
RMHNN(edge>14)(402)	93.10	88.85	93.57	90.46
RMHNN(scale=0.069)(402)	93.10	89.33	93.32	90.45

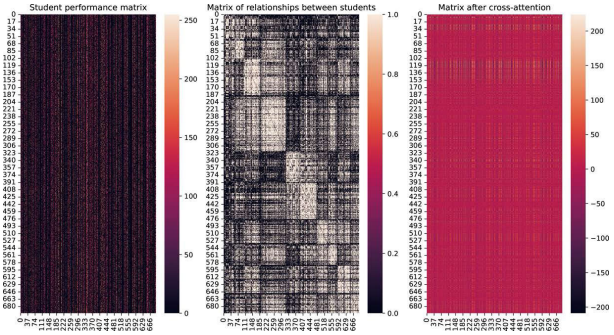


FIGURE 11. Embedding the student relation network in the graph convolutional layer and combining the grade matrix S with the student adjacency matrix in the cross-attention layer to visualize the model's prediction performance through heatmaps.

results show that the RMHNN model can effectively deal with the problem that the feature space is too large and the feature representation vectors are too sparse, which makes it difficult to capture the correlation of eigenvalues among students.

As shown in Figure 11 and Figure 12. Figure 11 illustrates the embedding of the student relation network into the graph convolutional layer to obtain a new graph representation. Subsequently, in the cross-attention layer, this novel graph representation is fused with the student adjacency matrix, followed by experimental predictions. This process is presented through visualization to showcase its impact on model predictions. Figure 12 demonstrates the embedding of the course relation network into the graph convolutional layer to obtain a new graph representation. Subsequently, in the cross-attention layer, this novel graph representation is fused with the course adjacency matrix, followed by experimental predictions. This process is presented through visualization to showcase its impact on model predictions. A comparison of the visualization effects reveals that the student relation network constructed based on the grade matrix S helps the model to capture the association of eigenvalues among student samples. By comparing the visualization results, it can be found that constructing the student relation network based on the grade matrix S helps the model to

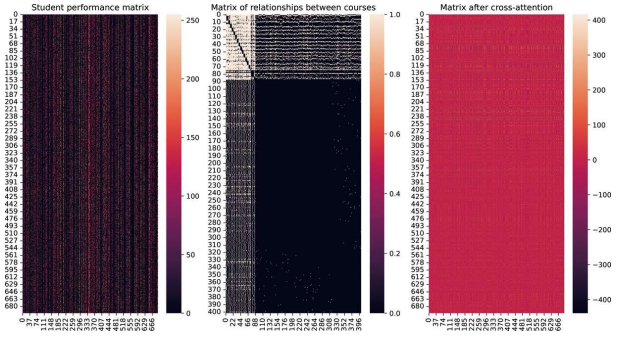


FIGURE 12. Embedding the course relation network in the graph convolution layer and combining the grade matrix S with the course adjacency matrix in the cross-attention layer, the model's predictive performance is visualized through heatmaps.

capture the correlation of eigenvalues among the student samples.

D. ANALYSIS OF EXPERIMENTAL RESULTS

In the task of assessing student performance, to address the difficulty of k-means clustering in dealing with high-dimensional and sparse discrete data matrices, we use a combination of the RMBN data processing approach and Louvain clustering to achieve an accurate assessment of student performance. The results of the experiment shows that the RMBN data processing approach enhances the connection between students and variables. Using a combination of the RMBN data processing approach and Louvain clustering, helps the algorithm to clarify the correlation between the students in the data matrix and to form groupings of students with a higher degree of modularity.

In the RMHNN model, we use GCN to embed the relation network into the grade matrix S, addressing the problem that using Euclidean distances to measure distances between discrete features is not practically meaningful. The experiments in this paper are carried out in the following seven areas. First, we apply the datasets to the GCN and GAT. By adjusting the adjacency matrices embedded into the models, we observe that altering the thresholds of the relationship edges could impact the accuracy of the models in predicting student performance. The experimental results also validate the effectiveness of the proposed rules for relation networks. In the GCN, the model achieves the best accuracy when the threshold of the relationship edge is set to be greater than 14. In the GAT, the accuracy of the model is best when the threshold of the relationship edge is set to be greater than 15. Second, to achieve finer weight adjustment and capture similar changes in eigenvalues between samples, this paper uses a linear function as the weight adjustment function to scale the data matrix. Experiments show that the use of linear scaling functions captures the correlation between samples and improves the model's ability to express complex relationships. In the GCN, the model has the best accuracy when scale=0.0695. In the GAT, the accuracy of the model is best when scale=0.075. Third, a potential issue with GCN is that they may suffer from feature-specificity problems. This paper conducts experiments to compare whether the introduction of the cross-attention layer has an impact on model performance. The experimental results show that the introduction of the cross-attention layer can effectively recognize the complex connection of eigenvalues between students and significantly improve the F1-score of the model. Fourth, normalization increases the stability of the model. The RMHNN model is used to evaluate the effect of normalized and unnormalized on the model performance with a relational edge threshold set. The experimental results indicate that, under the premise of setting the relationship edge threshold, data normalization is conducive to the model's learning and generalization. Fifth, under the premise of applying a linear scaling function, the RMHNN model is used to assess the impact of normalization and non-normalization on model performance. The experimental results indicate that applying normalization to the RMHNN model, which uses a linear scaling function as a weight adjustment function, can improve the accuracy and F1-score of the model in predicting student performance. Six, we conduct a comparative analysis of the proposed RMHNN model with traditional machine learning algorithms and neural network models, experimentally verifying the effectiveness of the RMHNN model. Seventh, the RMHNN model is subjected to ablation experiments. The results showed that changing the dimension of the features in the graph convolution layer within the RMHNN model has a moderate impact. Conversely, replacing the student-relational network embedded in the GCN with the course-relational network reduces the experimental effect.

VI. CONCLUSION

In the developmental process of student learning, predicting student performance can help educators gain a greater understanding of students' learning needs. To delve into students' academic achievements across various subjects and their practical performance in diverse scenarios, we conduct a comprehensive analysis of student data with multiple dimensions. This paper's approach not only centers on understanding the overall learning patterns exhibited by the majority of students but also considers the performance of each student across distinct subjects and their practical aptitude in varying situations. Therefore, we propose a student performance prediction approach. We apply the proposed model of student performance to real-world student data. The results demonstrate that our model is effectively predicting students' performance. By identifying students' performance, teachers can adjust their teaching strategies to cater to students' learning needs. By mastering the distribution of students' performance and future performance trends, educational administrators can deploy resources rationally to student groups with different performances.

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